

Kotak- IITM- Save Energy Mission

DETAILED ENERGY ASSESSMENT SUSTAINABILITY STUDY

BY IIT GANDHINAGAR

Company

Shree Mahadev Silk Mills Pvt. Ltd.

Address

Survey No. 142, Kadodara-Bardoli Road
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CLIENT LOGO

MAIN GATE / PLANT PHOTOGRAPH



Industrial Energy Assessment

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REPORT CONTROL

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ACKNOWLEDGEMENT

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The study team is indebted to Mr. Subhajit Bose of the plant for showing keen interest in the study and thankful to the M/s Shree Mahadev Silk Mills Pvt. Ltd. management for their wholehearted support and cooperation in the preparation of the energy audit report, without which the study would not have steered to its successful completion. Special thanks to the electrical maintenance team for their excellent coordination and seamless support for field measurement activities.

We also acknowledge all the transparency and courtesies extended during the stay and assessment activities.

It is well worth mentioning that the efforts being taken, and the enthusiasm shown by all the plant personnel towards energy conservation and sustainable growth were admirable. We found all the personnel keen to implement the possible energy conservation aspects.

Energy Assessor

IEA - IIT Gandhinagar

PREFACE

About Industrial Energy Assessment at IIT Gandhinagar

Industrial Energy Assessment Cell (IEAC) was established in the year 2022 in IC&SR, IIT Madras with the motto of "research and technological service to the nation towards sustainable energy and resource management". As a part of this, we conduct walk-through and detailed energy assessments in typical process industries to explore, evaluate and recommend hidden potentials for reducing their specific energy consumption and carbon emission in all possible ways.

IEAC have a team of professors from various domains to guide and evaluate our work. Our assessment team comprises of a Bureau of Energy Efficiency (BEE) - certified energy auditor and a couple of thermal experts to carry out energy assessments. The work described in this report was performed by the KISEM-IITGN Team in conjunction with IEAC, IIT Madras. This assessment is being carried out as a part of an industry outreach activity by IIT Gandhinagar Energy assessment team.



The objective of KISEM is to identify, evaluate, and recommend - through analyses of industrial plant operations - opportunities to conserve energy, minimize waste, and reduce the overall cost of operations. Our recommendations are based upon observations and measurements made at your plant. As our time was limited, we do not claim to have complete detail on every aspect of the plant's operations. At all times, we try to offer specific and quantitative recommendations of cost savings, energy conservation, and waste minimization of the plants. However, we do not attempt to prepare engineering designs or otherwise perform services that you would expect from an engineering firm, a vendor, or a manufacturer's representative. When the need for that kind of assistance arrives, we urge you to consult them directly.

For energy assessment case studies and the Energy Audit Tool, visit <https://kisem.org>

DISCLAIMER

The contents of this report are offered as guidance. All technical sources referenced in this report do not:

- Make any warranty or representation, expressed or implied, with respect to the accuracy, completeness, or usefulness of the information contained in this report, or that the use of any information, apparatus, method, or process disclosed in this report may not infringe on privately owned rights.
- The data and findings in the assessment activity are not publicly available and cannot be published on any platform.
- Assume any liabilities with respect to the use of, or for damages resulting from the use of, any information, apparatus, methods, or processes disclosed in this report. This report does not reflect the official views or policies of the above-mentioned institutions. Mention of trade names or commercial products does not constitute endorsement or recommendation of use.

Recommendations:

All the assessment recommendations are described in detail in the respective sections in this report. Further, the suggested actions are also indicated. The annual cost savings and implementation costs represent our best estimates. You may want to consult other sources to verify these estimates before making a final decision for implementing these recommendations. We welcome inquiries and further discussion on any information or data contained in this report.



CERTIFICATE OF ENERGY ASSESSMENT

This is to certify that M/s Shree Mahadev Silk Mills Pvt. Ltd. has participated in Industrial Energy assessment and Sustainability studies by IIT Gandhinagar. We appreciate the plant's efforts and support for facilitation of energy assessment and sustainability studies at plant premises. The data collection has been carried out diligently and truthfully; all data monitoring devices are in good working condition and have been calibrated or certified by approved agencies authorized and no tampering of such devices has occurred; all reasonable professional skill, care and diligence had been taken in preparing the energy audit report and the contents thereof are a true representation of the facts; adequate training provided to personnel involved in daily operations after implementation of recommendations; and the energy audit has been carried out in accordance with the Bureau of Energy Efficiency (Manner and Intervals of Time for the Conduct of Energy Audit) Regulations, 2010.

An overall summary of the energy assessment observations and recommendations described in this report is listed in the table on the following pages. The estimated total value of energy savings on an annualised basis in electrical energy is ₹ 15.36 Lac, which amounts to 4.6 % of the current electricity bill, and in thermal energy ₹ 18.34 Lac. Towards achieving these savings the estimated investment is ₹ 10.85 Lac, giving an overall return on investment in 3.9 Months. This also results in 703.80 tonnes of CO₂ emission reduction.

Rahul Jayantibhai Patel

CEA-30215, Lead GHG Verifier · Project Manager II

Industrial Energy Assessment Team – IIT Gandhinagar

DETAILS OF ENERGY ASSESSMENT TEAM

The following team was involved in the audit process for carrying out measurements, collection of data, and field study.

Plant team:

SR NO	NAME	DESIGNATION
1	Mr. Subhajit Bose	Manager – Utilities
2	Mr. Hitesh Patel	Electrical Supervisor

IEAC team – IIT Gandhinagar

SR NO	NAME	DESIGNATION
1	Dr. Naran Pindoriya	Project Coordinator, IEA – IIT Gandhinagar
2	Mr. Rahul Jayantibhai Patel	CEA-30215, Lead GHG Verifier Project Manager II



EXECUTIVE SUMMARY

ELECTRICAL SAVINGS**1,34,400 kWh****THERMAL SAVINGS****255 Tonne****EMISSIONS****703.80 tCO₂e****MONETARY****₹ 33.70 Lac****INVESTMENT****₹ 10.85 Lac****PAYBACK****3.9 Months**

An overall summary of the energy assessment observations and recommendations described in this report is listed in the table on the following pages. The estimated total value of energy savings on an annualised basis in electrical energy is ₹ 15.36 Lac, which amounts to 4.6 % of the current electricity bill, and in thermal energy ₹ 18.34 Lac. Towards achieving these savings the estimated investment is ₹ 10.85 Lac, giving an overall return on investment in 3.9 Months. This also results in 703.80 tonnes of CO₂ emission reduction.

All the recommendations are described in detail in the respective sections of this report. The annual cost savings and implementation costs represent our best estimates. The IITGN team welcomes inquiries and further discussion on any information or data contained in this report.

Observations and recommendations

SR	OBSERVATION	RECOMMENDATION	ANNUAL CONSUMPTION	UNIT	ANNUAL SAVING	% SAVING	MONETARY SAVING	INVESTMENT	ROI MONTHS
ELECTRICITY BILL									
1	Power factor fell below 0.95 in five of twelve months, the lowest being 0.936, forfeiting the utility rebate in those months.	Replace the failed capacitor element in APFC Stage 8 and add a relay-and-contactor set so the bank switches to hold the plant above 0.95 throughout.	32,87,000	kWh/yr	0	0.0	₹ 2.25 Lac	₹ 50,000/-	2.7 Months
2	Highest recorded maximum demand is 666 kVA against a contract demand of 800 kVA, so the plant pays the 680 kVA minimum billing demand every month.	Reduce contract demand from 800 kVA to 700 kVA. Minimum billing demand then becomes 595 kVA, closer to the actual load pattern, with adequate headroom retained.	—	kVA	0	—	₹ 1.53 Lac	₹ 0/-	Immediate
BOILER									



Table continued

SR	OBSERVATION	RECOMMENDATION	ANNUAL CONSUMPTION	UNIT	ANNUAL SAVING	% SAVING	MONETARY SAVING	INVESTMENT	ROI MONTHS
3	Flue gas oxygen of 10.55 % corresponds to about 101 % excess air, against the recommended 4–6 % O ₂ band for this boiler.	Tune the burner air-to-fuel ratio and install an online flue gas oxygen analyser for continuous combustion control.	1,138	tonne /yr	30	2.6	₹ 2.85 Lac	₹ 1.00 Lac	4.2 Months
JET MACHINES									
4	JET-01 is uninsulated with an average body temperature of 74.5 °C against 34 °C ambient; JET-03 is half insulated at 63.8 °C.	Insulate the shell of JET-01 and complete the insulation on JET-03, and replace the passing steam trap on JET-01.	1,138	tonne /yr	28	2.5	₹ 2.66 Lac	₹ 1.45 Lac	6.5 Months
5	10 of 14 jet machines (JET-01, JET-02, JET-03, JET-05, JET-06, JET-07, JET-08, JET-11 and others) are uninsulated or only half insulated. Measured surface temperatures give a combined recoverable steam loss of 32.4 kg/hr.	Provide standard LRB insulation on the bare and half-insulated jet bodies at ₹ 1,350 per Sq. Mtr.	—	tonne /yr	34	—	₹ 2.20 Lac	₹ 3.10 Lac	16.9 Months
6	6 circulation pump(s) — JET-01, JET-03, JET-05, JET-06, JET-12, JET-13 — measured below 40 % efficiency against an average of 46.0 % across the 14 machines surveyed.	Replace the below-benchmark jet circulation pumps with correctly sized units achieving at least 40 % efficiency at duty point.	—	kWh/yr	1,34,400	—	₹ 11.59 Lac	₹ 4.50 Lac	4.7 Months
7	3 steam trap(s) — JET-01, JET-03, JET-11 — were found passing live steam to drain.	Replace the passing steam traps at ₹ 10,000 each and put a quarterly trap survey in place.	—	tonne /yr	163	—	₹ 10.62 Lac	₹ 30,000/-	0.3 Months

PRODUCTION PROCESS AND DESCRIPTION

Plant introduction

Shree Mahadev Silk Mills is a textile processing unit engaged in dyeing and finishing of synthetic and blended fabrics.

The plant runs jet dyeing machines, a coal-fired boiler and a thermic fluid heater supplying the stenter line, with a connected load of approximately 1.2 MW.

Products:

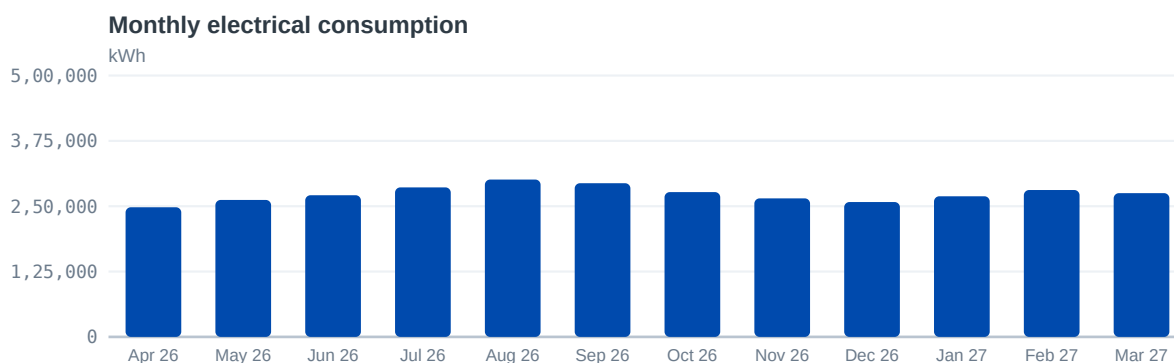


SR. NO.	PRODUCT	CAS / GRADE	STATUS
1	Dyed polyester fabric	—	Commercial
2	Finished blended fabric	—	Commercial

BASELINE OF PLANT ENERGY CONSUMPTION

Baseline of plant energy consumption – electrical

MONTH	KWH	TOE
Apr 26	2,48,000	21.323
May 26	2,62,000	22.527
Jun 26	2,71,000	23.301
Jul 26	2,86,000	24.590
Aug 26	3,01,000	25.880
Sep 26	2,94,000	25.278
Oct 26	2,77,000	23.816
Nov 26	2,65,000	22.785
Dec 26	2,58,000	22.183
Jan 27	2,69,000	23.129
Feb 27	2,81,000	24.160
Mar 27	2,75,000	23.645
Total	32,87,000	282.616

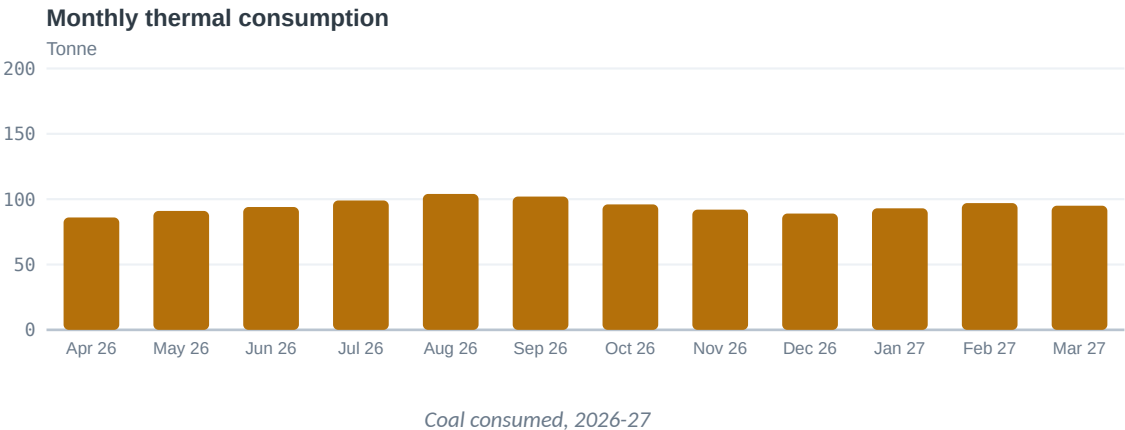


Purchased units, 2026-27



Baseline of plant energy consumption – thermal

MONTH	COAL (TONNE)	TOE
Apr 26	86.00	0.036
May 26	91.00	0.038
Jun 26	94.00	0.039
Jul 26	99.00	0.042
Aug 26	104.00	0.044
Sep 26	102.00	0.043
Oct 26	96.00	0.040
Nov 26	92.00	0.039
Dec 26	89.00	0.037
Jan 27	93.00	0.039
Feb 27	97.00	0.041
Mar 27	95.00	0.040



GHG EMISSION ACCOUNTING

Boundaries of the emissions accounting

Emissions accounting covers Scope 1 (direct, stationary combustion) and Scope 2 (indirect, purchased electricity). Scope 3 activity data was not available for this assessment and is excluded.

Scope 2 uses CEA CO2 Baseline Database v21.0, Dec 2025. Scope 1 uses UK DEFRA GHG conversion factors 2025. Both factor versions are pinned into this report.



Summary of GHG emissions

SCOPE	TYPE OF EMISSION	SOURCE	EMISSION TCO ₂ E
Scope 1	Stationary combustion	Coal	2708.44
Scope 2	Purchased electricity	Grid	2353.49
Total			5061.93

ELECTRICITY BILL ANALYSIS

Plant electricity bill analysis

MONTH	CD KVA	ACTUAL MD	BILLING DMD	KWH	KVAH	PF	NET ₹
Apr 25	800	666	680	3,98,588	4,05,069	0.984	30,87,179
May 25	800	624	680	4,14,193	4,34,164	0.954	31,95,842
Jun 25	800	587	680	2,90,978	3,02,157	0.963	23,20,524
Jul 25	800	526	680	4,18,511	4,53,424	0.923	32,27,437
Aug 25	800	629	680	3,31,202	3,46,083	0.957	26,06,559
Sep 25	800	663	680	4,24,511	4,27,073	0.994	32,71,952
Oct 25	800	538	680	2,98,408	3,10,518	0.961	23,76,061
Nov 25	800	678	680	2,80,142	2,99,937	0.934	22,45,606
Dec 25	800	652	680	4,28,436	4,43,515	0.966	33,00,344
Jan 26	800	640	680	3,10,930	3,35,053	0.928	24,62,946
Feb 26	800	552	680	3,29,384	3,45,991	0.952	25,99,242
Mar 26	800	520	680	3,90,298	3,93,049	0.993	30,23,978

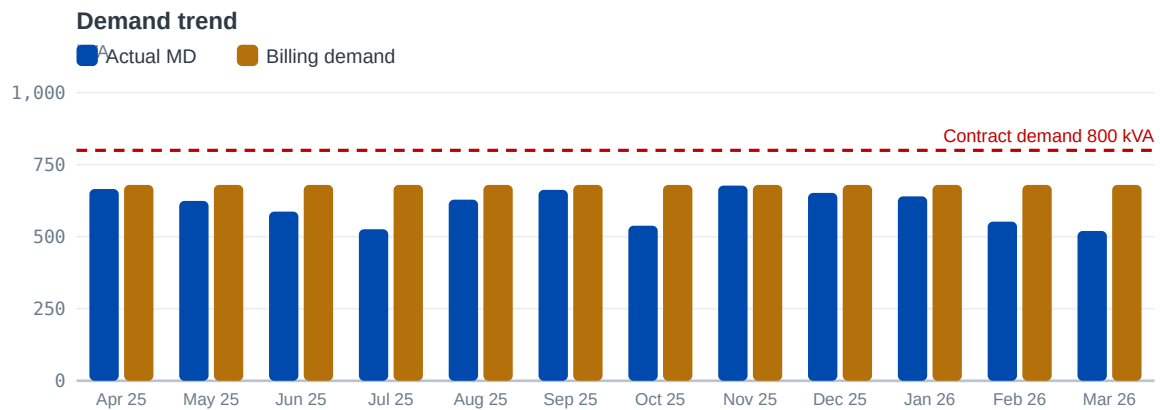
Observations in electricity bills

Average units consumed = 3,59,632 units/month
Average actual maximum demand = 606.33 kVA
Average power factor = 0.959
Contract demand = 800 kVA
Load factor = $\text{avg kWh} / (\text{avg MD} \times \text{avg PF} \times 24 \times 30) = 0.859$
Plant utility = $\text{avg kWh} / (\text{CD} \times \text{avg PF} \times 24 \times 30) = 0.651$
Demand factor = $\text{avg actual MD} / \text{CD} = 0.758$

The higher the load factor, the better the utilisation of the equipment and installed capacity.

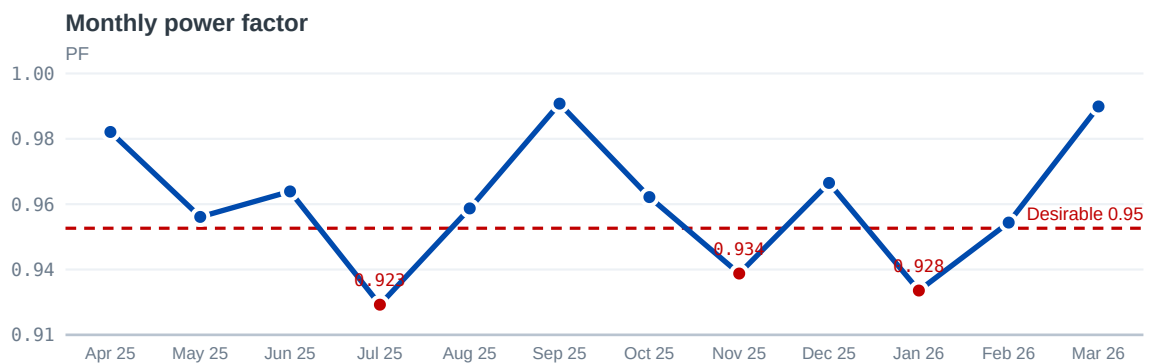


Contract demand vs actual and billing demand



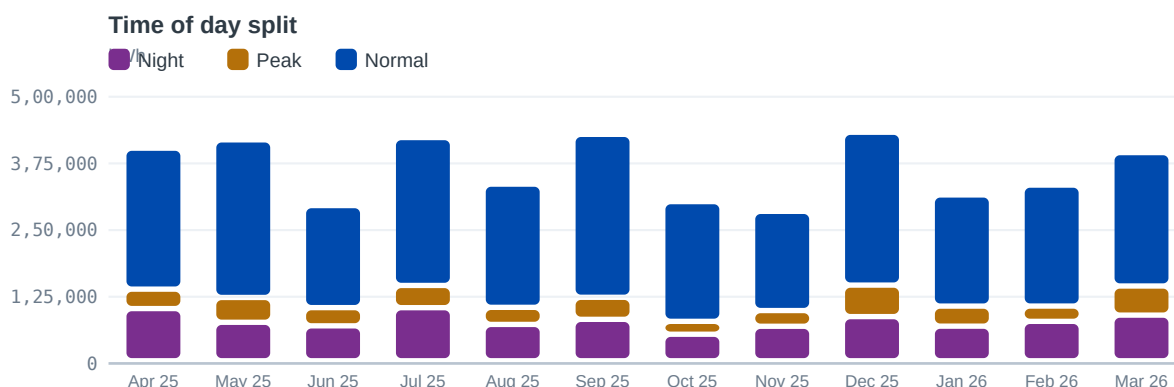
Actual maximum demand against billing demand, with the sanctioned contract demand marked.

Annual power factor trend



Months below 0.95 are marked and labelled — those are the months carrying a penalty.

Time of day (TOD) in electricity bill



Night-hour units attract a rebate; peak-hour units a surcharge.

Recommendation in electricity bills

The monthly power factor dropped below the desirable level of 0.95 in five months, with the lowest recorded value 0.936. A low power factor forfeits the utility rebate and increases reactive current, system losses and transformer loading.

The plant operates at a contract demand of 800 kVA against a highest recorded maximum demand of 666 kVA. Because minimum billing demand is 85 % of contract demand, the plant pays for 680 kVA every month regardless of use.

Highest recorded maximum demand is 678 kVA against a contract demand of 800 kVA — 15 % headroom.
 Contract demand optimisation is worth evaluating, remembering that billing demand will not fall below 85 % of whatever contract demand is set.

1. Replace the failed capacitor element in APFC Stage 8 and add a relay-and-contactor set so the bank switches to hold the plant above 0.95 throughout.

Power factor fell below 0.95 in five of twelve months, the lowest being 0.936, forfeiting the utility rebate in those months.

Annual energy saving = 0 kWh/yr
 Annual monetary saving = ₹ 2.25 Lac
 Investment = ₹ 50,000/-
 Simple payback period = 2.7 Months

2. Reduce contract demand from 800 kVA to 700 kVA. Minimum billing demand then becomes 595 kVA, closer to the actual load pattern, with adequate headroom retained.



Highest recorded maximum demand is 666 kVA against a contract demand of 800 kVA, so the plant pays the 680 kVA minimum billing demand every month.

Annual energy saving = 0 kVA
Annual monetary saving = ₹ 1.53 Lac
Investment = ₹ 0/-
Simple payback period = Immediate

ASSESSMENT OF ELECTRICAL DISTRIBUTION SYSTEM

Plant load demand study

MAINLINE DEMAND MONITORING	CONTRACT DEMAND	AVERAGE	MINIMUM	MAXIMUM
Demand kVA	800	508.91	98.26	607.11

Recording window: 9th June 2026 10:06 to 10th June 2026 01:52 (27 hrs recording)

Plant section-wise energy consumption — electrical PCC load

NAME OF MACHINE	VOLTAGE	CURRENT	KW	Q KVAR	KVA	PF	%V THD	%I THD
New Panel Power	430.06	485.38	351.15	-11.14	366.04	0.959	6.01	22.90
Old Panel Power	428.40	229.10	167.60	17.35	170.39	0.983	3.15	8.37
Stenter MCC	429.10	141.20	98.40	22.10	104.85	0.938	5.80	18.40

Red cells exceed the IEEE-519:2022 limits of 5 % voltage THD and 8 % current THD.

Motor load study

S.N O	NAME	RATED KW	START ER	FREQ /RPM	VOLTA GE	CURR ENT	KW	KVAR	KVA	PF	% LOAD
1	Jet Dyeing Pump 1	37.0	VFD	44	415	52.1	33.80	9.10	37.40	0.904	91.4
2	Jet Dyeing Pump 2	37.0	VFD	46	415	58.4	39.10	10.40	41.90	0.933	105.7
3	Stenter Exhaust Fan	22.0	DOL	50	414	29.8	16.40	8.90	21.40	0.766	74.5
4	Boiler FD Fan	15.0	DOL	50	416	20.1	11.20	6.10	14.50	0.772	74.7
5	Cooling Tower Pump	18.5	DOL	50	415	24.9	8.90	6.80	17.90	0.497	48.1

Above 100 % is an overloaded motor; below 50 % is an oversized one. The recommended band is 70–90 %.



Jet Dyeing Pump 2 runs at 105.7 % of rated capacity. Continuous operation above rating raises winding temperature, degrades insulation and shortens motor life. The cooling tower pump at 48 % load is oversized for its duty.

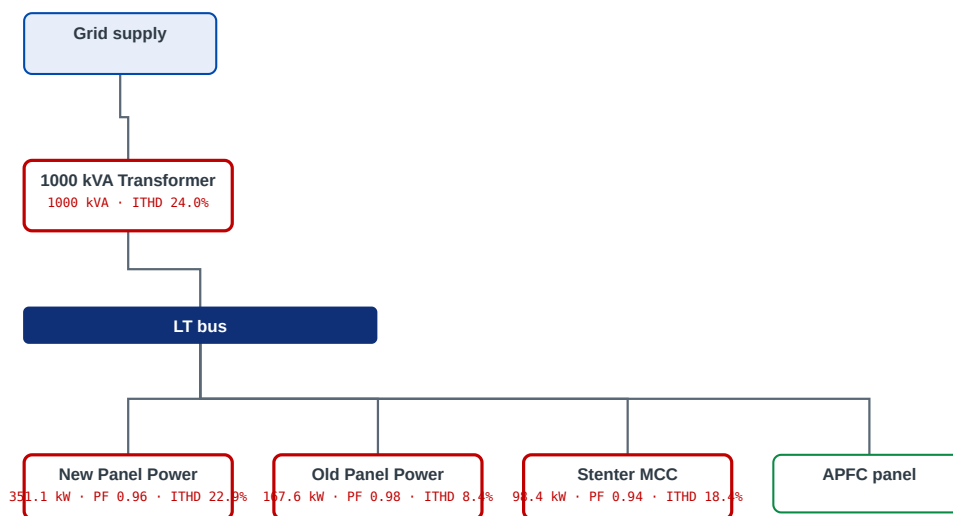
Automatic power factor correction study

STAGE	RATED KVAR	VOLTAGE	I-R	I-Y	I-B	REMARK
Stage 1	25.0	415	34.2	34.0	33.8	Healthy
Stage 2	25.0	415	33.9	34.1	34.0	Healthy
Stage 8	50.0	415	41.2	40.8	22.1	Derated — one phase weak

Stage 8 draws markedly lower current on one phase, indicating a failed capacitor element. Effective bank capacity is therefore below nameplate.

POWER QUALITY AND TRANSFORMER ASSESSMENT

Plant single line diagram



Generated from the panel hierarchy. Nodes breaching IEEE-519 are outlined in red; dashed nodes are provisional, recorded at the walkthrough and not yet confirmed.



Rated details of transformer

Make	VOLTAMP	Capacity	1,000 kVA
Primary voltage	11000 V	Secondary voltage	433 V
Impedance	5.02 %	Year	2016
No-load loss	1.35 kW	Load loss	9.80 kW
Oil quantity	980 litre	Serial no.	VT-2016-4471

Transformer efficiency monitoring

Average loading	38.2 %
Standard efficiency at this loading	99.28 %
Actual load-based efficiency	99.02 %

Power quality as per IEEE-519:2022 — harmonics

PARAMETER	RECOMMENDED LIMIT	MEASURED	VERDICT
Voltage THD (%VTHD)	< 5.00 %	5.61 %	Exceeds limit
Current THD (%ITHD)	< 8.00 %	24.01 %	Exceeds limit

Harmonics mitigation action plan

Measured harmonic levels are above the IEEE-519 recommended limits, which contributes to transformer overheating, additional system losses and reduced equipment life. The usual measures, in order of priority:

MEASURE	APPLIES TO	BENEFIT	PRIORITY
Install 7 % detuned reactors with the APFC bank	Main PCC APFC panel	Prevents resonance and protects the capacitor bank	Critical
Active harmonic filter at the PCC	Main incomer	Cuts current THD at source	High
Segregate non-linear loads onto their own feeders	VFD-driven drives	Reduces harmonic propagation across the PCC	Medium

Transformer thermography

No critical temperature was found on the transformer body, bushings or the gantry structure. Maximum recorded surface temperature was 52.4 °C against an ambient of 34 °C.

New Panel Main Line Transformer

Recording ID: REC-0612-A



PARAMETER	AVERAGE	MIN	MAX
Apparent power kVA	366.04	307.53	423.75
Active power kW	351.15	297.52	404.06
Power factor	0.96	0.89	0.97
Voltage THD %	6.01	5.12	7.55
Current THD %	23.84	18.10	33.20

PERFORMANCE ASSESSMENT OF MAJOR PLANT UTILITIES

Performance assessment of boiler

Boiler efficiency is the percentage of heat input that is effectively utilised to generate steam. There are two methods of assessing it.

Direct method (input-output method): the energy gain of the working fluid is compared with the energy content of the fuel. It needs only the useful output (steam) and the heat input (fuel).

Indirect method (heat loss method): efficiency is 100 minus the sum of the losses - dry flue gas, moisture in fuel and combustion air, combustion of hydrogen, radiation, and unburnt.

Technical specification

PARAMETER	VALUE
Boiler ID and make	GT-6942, Thermax
Type	Solid fuel fired, water tube
Rated capacity	4 TPH
Design pressure	10.54 kg/cm ²
Fuel	Indian coal

Efficiency — direct method

PARAMETER	VALUE	UNIT
Steam generated	3450.0	kg/hr
Feed water temperature	82.0	°C
Steam enthalpy	664.0	kcal/kg
Fuel consumed	790.00	kg/hr
GCV of fuel	4200	kcal/kg



Table continued

PARAMETER	VALUE	UNIT
Evaporation ratio	4.37	—
Efficiency (direct)	60.5 %	%

Efficiency — indirect method

FLUE GAS ANALYSIS	VALUE	UNIT
Oxygen in flue gas	10.55	%
Carbon monoxide	0	ppm
Flue gas temperature	185.3	°C
Ambient temperature	34.0	°C
Excess air	101.0	%
Dry flue gas loss	19.32	%
Radiation & unaccounted loss	1.50	%
Efficiency (indirect)	79.2 %	%

Oxygen of 10.55 % corresponds to roughly 101 % excess air, well above the 4–6 % O₂ band for efficient combustion. Excess air raises dry flue gas losses directly.

CO of 0 ppm indicates complete combustion, so there is no incomplete-combustion loss to recover.

Average oxygen in the flue gas was 10.55 %, corresponding to roughly 101 % excess air, well above the recommended range. CO was 0 ppm throughout, so combustion is complete and the loss is entirely in excess air.

Recommendation in boiler

1. Tune the burner air-to-fuel ratio and install an online flue gas oxygen analyser for continuous combustion control.

Flue gas oxygen of 10.55 % corresponds to about 101 % excess air, against the recommended 4–6 % O₂ band for this boiler.

Annual energy saving = 30 tonne/yr
Annual monetary saving = ₹ 2.85 Lac
Investment = ₹ 1.00 Lac
Simple payback period = 4.2 Months



Performance assessment of thermic oil heater

Boiler efficiency is the percentage of heat input that is effectively utilised to generate steam. There are two methods of assessing it.

Direct method (input-output method): the energy gain of the working fluid is compared with the energy content of the fuel. It needs only the useful output (steam) and the heat input (fuel).

Indirect method (heat loss method): efficiency is 100 minus the sum of the losses - dry flue gas, moisture in fuel and combustion air, combustion of hydrogen, radiation, and unburnt.

Pumping system analysis

Pumps and blowers operate at optimum efficiency only at their rated head and output; at other operating conditions efficiency falls. Where variable flow is required, variable speed drives reduce power consumption roughly in proportion to the flow requirement, following the affinity laws.

Three parameters are needed to determine pump efficiency: flow, head and power. Flow is the hardest - where no permanent flow meter exists, a portable ultrasonic meter measures it non-invasively.

PUMP	MAKE	RATED KW	FLOW M ³ /HR	HEAD M	MOTOR KW	HYDRAULIC KW	SHAFT KW	PUMP EFF %
Boiler feed pump	Kirloskar	15.0	11.2	110.0	12.80	3.36	11.65	28.8
Cooling tower pump	CRI	18.5	148.0	18.0	8.90	7.26	7.92	91.6

Pump efficiency below 50 % usually justifies replacement; 50–65 % justifies impeller trimming or a VFD.

Performance assessment of jet machine

Methodology followed for jet analysis

Plant has installed 14 Nos. of steam jet dyeing machines in the plant premises. A detailed thermal study of the jet was performed on each individual jet in its heating state and the losses were derived. The following measurements were taken on every machine:



- Jet machine water-side flow measurement
- Jet pressure measurement
- Circulation pump efficiency
- Pump power consumption
- Surface heat loss from the jet body
- Heat exchanger performance of the jet
- Steam and water temperature across the jet heat exchanger
- Steam trap performance assessment

Basis of calculation

PARAMETER	SYMBOL / BASIS	VALUE
Evaporation ratio of fuel	Steam generated per kg of fuel	4.01 kg/kg
Fuel cost	Coal, delivered	₹ 6.50 /kg
Electricity unit rate	Landed cost per kWh	₹ 8.62 /kWh
Insulation cost	Standard LRB insulation	₹ 1,350 per Sq. Mtr
Jet pump investment	Replacement, per pump	₹ 75,000 /-
Steam trap replacement	Per faulty trap	₹ 10,000 /-
Annual operating days	Working days considered	350 days

Insulation and trap savings are computed on each machine's own recorded heating hours per day; pump savings on its recorded running hours per day. Pump investment is counted only where measured efficiency is below 40 %, and trap investment only where the trap is passing or the drain sits before the trap — so no machine carries an investment for a fault it does not have.

Jet analysis summary

JETS SURVEYED

14

STEAM SAVING

32.4 kg/hr

TOTAL SAVING / YEAR

₹ 24.41 Lac

AVG PUMP EFFICIENCY

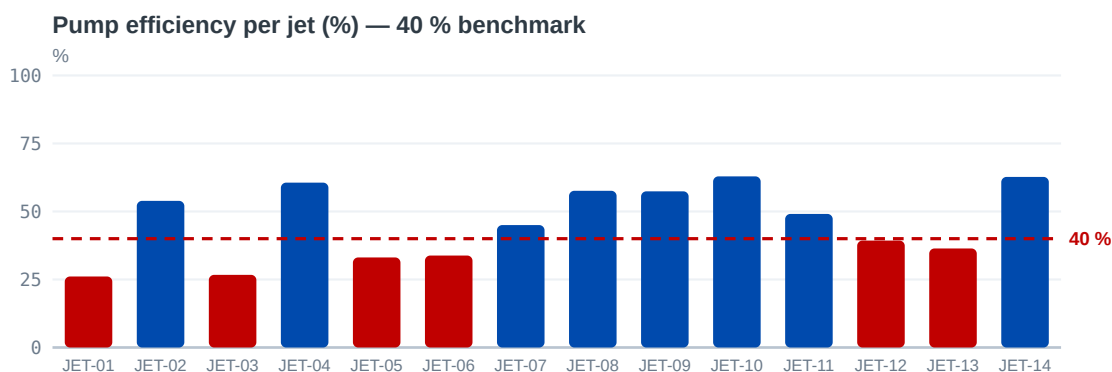
46.0 %

3 steam traps are passing live steam to drain — attend on JET-01, JET-03, JET-11.

6 circulation pumps are operating below 40 % efficiency — replace on JET-01, JET-03, JET-05, JET-06, JET-12, JET-13.



Steam recoverable on each machine by bringing the uninsulated body up to standard LRB insulation.



Bars below the dashed line are the pumps carrying a replacement recommendation.

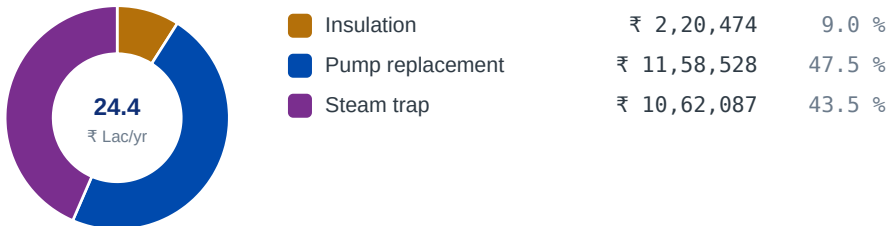
Operation mix at the time of survey



The state each machine was found in — heat-exchange performance is judged against the branch that applies.



Annual saving breakdown (₹/yr)



Where the money is. Investment across all three streams is ₹ 7.90 Lac, giving a blended payback of 3.9 Months.

Jets overview

JET NO.	OPERATION	PUMP EFF. %	HEAT EXCHANGE %	TRAP STATUS	STEAM SAVING KG/HR	SAVING ₹/YR	BEST ROI
JET-01	Heating	26.1	79.3	Trap Passing	3.00	6,42,362	0.3 Months
JET-02	Heating	53.9	85.0	Working OK	0.26	1,775	89.2 Months
JET-03	Heating	26.7	55.8	Trap Passing	4.90	7,25,894	0.3 Months
JET-04	Heating	60.6	52.5	Working OK	0.00	0	—
JET-05	Heating	33.1	57.8	Working OK	0.00	1,43,609	6.3 Months
JET-06	Heating	33.8	53.3	Working OK	1.90	2,59,129	3.7 Months
JET-07	Heating	45.0	57.3	Working OK	7.40	50,369	6.2 Months
JET-08	Heating	57.6	60.2	Working OK	2.49	16,952	11.2 Months
JET-09	Cooling	57.4	86.6	Working OK	0.00	0	—
JET-10	Cooling	62.9	85.7	Working OK	0.00	0	—
JET-11	Cooling	49.1	53.9	Trap Passing	5.62	3,92,275	0.3 Months
JET-12	Hold	39.3	51.7	Working OK	0.15	32,384	28.7 Months
JET-13	Hold	36.4	71.3	Working OK	6.67	1,76,340	6.9 Months
JET-14	Hold	62.7	68.1	Working OK	0.00	0	—
Total		46.0 %		3 passing	32.4	24,41,089	3.9 Months

Performance assessment of jet

Each machine is reported against the same spine of measurements, five machines to a table. A. insulation, B. pump and C. steam trap are the three independent saving streams; the financial block at the foot of each column adds only the streams that machine actually earns.

**Jets 1–5 of 14**

PARAMETER	JET-01	JET-02	JET-03	JET-04	JET-05
JET OPERATION DATA					
Capacity, kg	300	600	500	300	500
Process	Heating	Heating	Heating	Heating	Heating
Surface area, Sq. Mtr	21.32	9.77	15.16	17.56	16.98
Jet set temperature, °C	97.8	102.2	108.9	117.7	99.6
Insulation condition	UnInsulated	Half Insulated	UnInsulated	Insulated	UnInsulated
JET SURFACE HEAT LOSS					
Average body temperature, °C	55.4	47.1	67.2	36.1	41.1
Ambient temperature, °C	34.0	34.0	34.0	34.0	34.0
Surface heat loss, kCal/hr	2,627	651	3,239	119	526
Steam pressure, kg/cm ²	3.6	4.2	3.8	4.3	3.9
Eq. steam loss, kg/hr	5.31	1.32	6.54	0.24	1.06
Eq. fuel loss, kg/hr	1.324	0.329	1.631	0.060	0.264
After insulation body temp, °C	45.0	45.0	45.0	36.1	45.0
Eq. fuel loss after, kg/hr	0.576	0.264	0.409	0.060	0.459
Savings in coal, kg/hr	0.748	0.065	1.222	0.000	0.000
Annual fuel saving, Tonne	3.14	0.27	5.13	0.00	0.00
A. Annual savings, ₹*	20,423	1,775	33,358	0	0
JET PUMP PERFORMANCE					
Jet flow, CMH	138.1	276.0	308.1	310.6	217.6
Pump pressure, kg/cm ²	4.07	2.75	3.53	3.29	3.20
Power input to jet, kW	12.79	25.77	16.88	15.99	13.82
Hydraulic power, kW	3.34	13.89	4.51	9.69	4.57
Pump efficiency, %	26.1	53.9	26.7	60.6	33.1
After replacement, kW	8.35	—	11.27	—	11.44
Savings at min. 40 % eff., kW	4.44	—	5.61	—	2.38
Annual savings in units, kWh*	31,080	—	39,270	—	16,660
B. Savings in ₹	2,67,910	—	3,38,507	—	1,43,609
STEAM TRAP PERFORMANCE					
Trap type	Thermodynamic Trap	Float Type Trap	Float Type Trap	Float Type Trap	Float Type Trap
Trap temperature in, °C	144.3	146.3	148.2	145.9	142.2
Trap temperature out, °C	92.9	137.9	133.6	111.5	114.9
Trap working	Trap Passing	Working OK	Trap Passing	Working OK	Working OK
Loss of steam trap, kg/hr	52.00	4.59	52.00	0.37	4.55

**Jets 1–5 of 14 (contd.)**

PARAMETER	JET-01	JET-02	JET-03	JET-04	JET-05
C. Annual savings in ₹*	3,54,029	31,259	3,54,029	2,509	30,986
HEAT EXCHANGER PERFORMANCE					
Estd. / meas. steam flow, kg/hr	225.79	394.42	250.93	251.09	347.18
Steam / cooling water inlet, °C	152.4	152.7	156.5	145.7	151.7
Steam / cooling water outlet, °C	103.5	121.1	138.1	95.0	135.9
Water inlet, °C	71.4	71.5	77.3	61.3	66.0
Water outlet, °C	111.8	121.3	103.4	101.3	114.3
Jet heat exchange performance, %	79.3	85.0	55.8	52.5	57.8
FINANCIAL CALCULATION					
Total savings (A+B+C), ₹	6,42,362	1,775	7,25,894	0	1,43,609
Total investment, ₹	1,13,782	13,190	1,05,466	23,706	97,923
ROI in months	2.1 Months	89.2 Months	1.7 Months	—	8.2 Months

* Annual savings computed on the machine's own recorded operating hours and 350 working days.

Jets 6–10 of 14

PARAMETER	JET-06	JET-07	JET-08	JET-09	JET-10
JET OPERATION DATA					
Capacity, kg	300	250	300	500	300
Process	Heating	Heating	Heating	Cooling	Cooling
Surface area, Sq. Mtr	19.13	19.17	11.77	15.60	18.71
Jet set temperature, °C	124.2	110.8	130.9	114.1	111.2
Insulation condition	Half Insulated	UnInsulated	UnInsulated	Insulated	Insulated
JET SURFACE HEAT LOSS					
Average body temperature, °C	52.5	71.0	60.1	41.1	36.8
Ambient temperature, °C	34.0	34.0	34.0	34.0	34.0
Surface heat loss, kCal/hr	1,964	4,692	1,860	484	181
Steam pressure, kg/cm ²	3.7	5.5	4.7	4.4	5.4
Eq. steam loss, kg/hr	3.97	9.48	3.76	0.98	0.37
Eq. fuel loss, kg/hr	0.990	2.364	0.938	0.244	0.092
After insulation body temp, °C	45.0	45.0	45.0	41.1	36.8
Eq. fuel loss after, kg/hr	0.516	0.519	0.317	0.244	0.092
Savings in coal, kg/hr	0.474	1.845	0.621	0.000	0.000
Annual fuel saving, Tonne	1.99	7.75	2.61	0.00	0.00
A. Annual savings, ₹*	12,942	50,369	16,952	0	0
JET PUMP PERFORMANCE					

**Jets 6–10 of 14 (contd.)**

PARAMETER	JET-06	JET-07	JET-08	JET-09	JET-10
Jet flow, CMH	317.9	200.3	252.5	308.3	204.3
Pump pressure, kg/cm ²	3.18	3.81	3.29	4.19	2.23
Power input to jet, kW	26.32	19.24	15.93	19.42	29.81
Hydraulic power, kW	8.90	8.66	9.18	11.15	18.75
Pump efficiency, %	33.8	45.0	57.6	57.4	62.9
After replacement, kW	22.24	—	—	—	—
Savings at min. 40 % eff., kW	4.08	—	—	—	—
Annual savings in units, kWh*	28,560	—	—	—	—
B. Savings in ₹	2,46,187	—	—	—	—
STEAM TRAP PERFORMANCE					
Trap type	Thermostatic Trap	Float Type Trap	Thermostatic Trap	Thermostatic Trap	Thermostatic Trap
Trap temperature in, °C	143.4	142.3	142.6	150.1	154.8
Trap temperature out, °C	95.8	115.2	111.0	108.6	128.4
Trap working	Working OK	Working OK	Working OK	Working OK	Working OK
Loss of steam trap, kg/hr	2.90	0.09	3.34	4.02	0.50
C. Annual savings in ₹*	19,741	598	22,744	27,352	3,413
HEAT EXCHANGER PERFORMANCE					
Estd. / meas. steam flow, kg/hr	572.28	568.29	296.79	558.06	577.89
Steam / cooling water inlet, °C	149.5	150.6	146.4	32.0	32.9
Steam / cooling water outlet, °C	123.9	134.2	120.2	41.7	50.2
Water inlet, °C	79.7	79.4	66.5	75.7	65.4
Water outlet, °C	113.3	114.5	110.6	121.9	122.2
Jet heat exchange performance, %	53.3	57.3	60.2	86.6	85.7
FINANCIAL CALCULATION					
Total savings (A+B+C), ₹	2,59,129	50,369	16,952	0	0
Total investment, ₹	1,00,826	25,880	15,890	21,060	25,259
ROI in months	4.7 Months	6.2 Months	11.2 Months	—	—

* Annual savings computed on the machine's own recorded operating hours and 350 working days.

Jets 11–14 of 14

PARAMETER	JET-11	JET-12	JET-13	JET-14
JET OPERATION DATA				
Capacity, kg	250	250	600	300
Process	Cooling	Hold	Hold	Hold
Surface area, Sq. Mtr	10.94	15.50	20.32	17.55

**Jets 11-14 of 14 (contd.)**

PARAMETER	JET-11	JET-12	JET-13	JET-14
Jet set temperature, °C	130.8	134.1	95.1	100.8
Insulation condition	UnInsulated	UnInsulated	UnInsulated	Insulated
JET SURFACE HEAT LOSS				
Average body temperature, °C	78.4	45.8	67.5	42.4
Ambient temperature, °C	34.0	34.0	34.0	34.0
Surface heat loss, kCal/hr	3,367	907	4,390	671
Steam pressure, kg/cm ²	4.3	3.9	4.1	3.5
Eq. steam loss, kg/hr	6.80	1.83	8.87	1.36
Eq. fuel loss, kg/hr	1.696	0.456	2.212	0.339
After insulation body temp, °C	45.0	45.0	45.0	42.4
Eq. fuel loss after, kg/hr	0.294	0.419	0.549	0.339
Savings in coal, kg/hr	1.401	0.037	1.663	0.000
Annual fuel saving, Tonne	5.88	0.15	6.99	0.00
A. Annual savings, ₹*	38,246	1,008	45,403	0
JET PUMP PERFORMANCE				
Jet flow, CMH	173.8	222.8	192.8	285.0
Pump pressure, kg/cm ²	2.22	3.12	4.46	3.28
Power input to jet, kW	17.21	29.90	24.16	25.52
Hydraulic power, kW	8.45	11.75	8.79	16.00
Pump efficiency, %	49.1	39.3	36.4	62.7
After replacement, kW	—	29.38	21.99	—
Savings at min. 40 % eff., kW	—	0.52	2.17	—
Annual savings in units, kWh*	—	3,640	15,190	—
B. Savings in ₹	—	31,377	1,30,938	—
STEAM TRAP PERFORMANCE				
Trap type	Float Type Trap	Thermodynamic Trap	Thermodynamic Trap	Thermodynamic Trap
Trap temperature in, °C	149.3	145.1	143.7	141.3
Trap temperature out, °C	87.4	130.8	127.7	87.3
Trap working	Trap Passing	Working OK	Working OK	Working OK
Loss of steam trap, kg/hr	52.00	2.43	1.59	5.76
C. Annual savings in ₹*	3,54,029	16,543	10,836	39,202
HEAT EXCHANGER PERFORMANCE				
Estd. / meas. steam flow, kg/hr	290.57	256.73	455.22	303.98
Steam / cooling water inlet, °C	33.6	154.6	149.0	151.9
Steam / cooling water outlet, °C	47.5	106.5	105.5	106.0



Jets 11–14 of 14 (contd.) (contd.)

PARAMETER	JET-11	JET-12	JET-13	JET-14
Water inlet, °C	78.8	67.0	61.8	67.5
Water outlet, °C	124.1	96.6	107.0	108.5
Jet heat exchange performance, %	53.9	51.7	71.3	68.1
FINANCIAL CALCULATION				
Total savings (A+B+C), ₹	3,92,275	32,384	1,76,340	0
Total investment, ₹	24,769	95,925	1,02,432	23,693
ROI in months	0.8 Months	35.5 Months	7.0 Months	—

* Annual savings computed on the machine's own recorded operating hours and 350 working days.

Observations recorded at the machine

JET NO.	OBSERVATION
JET-08	Body insulation damaged near manhole.
JET-12	Body insulation damaged near manhole.

Recommendation in jet machines

1. Insulate the shell of JET-01 and complete the insulation on JET-03, and replace the passing steam trap on JET-01.

JET-01 is uninsulated with an average body temperature of 74.5 °C against 34 °C ambient; JET-03 is half insulated at 63.8 °C.

Annual energy saving = 28 tonne/yr
 Annual monetary saving = ₹ 2.66 Lac
 Investment = ₹ 1.45 Lac
 Simple payback period = 6.5 Months

2. Provide standard LRB insulation on the bare and half-insulated jet bodies at ₹ 1,350 per Sq. Mtr.

10 of 14 jet machines (JET-01, JET-02, JET-03, JET-05, JET-06, JET-07, JET-08, JET-11 and others) are uninsulated or only half insulated. Measured surface temperatures give a combined recoverable steam loss of 32.4 kg/hr.

Annual energy saving = 34 tonne/yr
 Annual monetary saving = ₹ 2.20 Lac
 Investment = ₹ 3.10 Lac
 Simple payback period = 16.9 Months

3. Replace the below-benchmark jet circulation pumps with correctly sized units achieving at least 40 % efficiency at duty point.



6 circulation pump(s) — JET-01, JET-03, JET-05, JET-06, JET-12, JET-13 — measured below 40 % efficiency against an average of 46.0 % across the 14 machines surveyed.

Annual energy saving = 1,34,400 kWh/yr
Annual monetary saving = ₹ 11.59 Lac
Investment = ₹ 4.50 Lac
Simple payback period = 4.7 Months

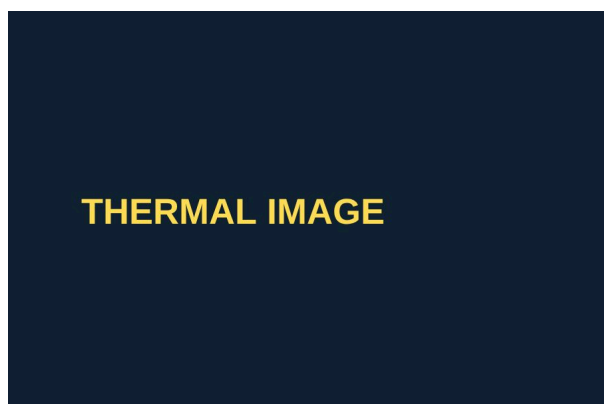
4. Replace the passing steam traps at ₹ 10,000 each and put a quarterly trap survey in place.

3 steam trap(s) — JET-01, JET-03, JET-11 — were found passing live steam to drain.

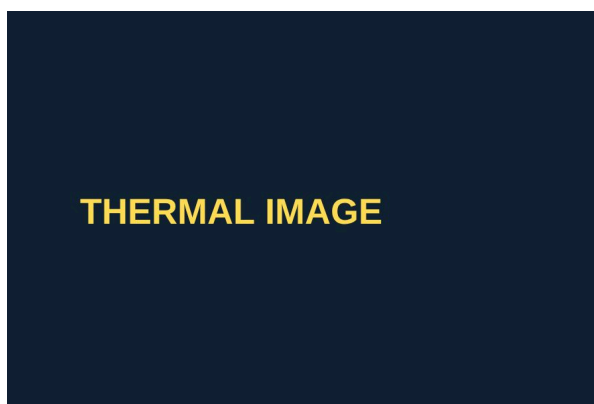
Annual energy saving = 163 tonne/yr
Annual monetary saving = ₹ 10.62 Lac
Investment = ₹ 30,000/-
Simple payback period = 0.3 Months

THERMAL IMAGING SURVEY

Thermography was carried out on all jet bodies during the heating cycle.



JET-01 shell, 74.5 C



JET-03 manhole, 63.8 C

LOCATION	MEASURED C	LIMIT C
JET-01 shell	74.5	45
JET-03 manhole	63.8	45

Two surfaces exceed the 45 C touch-safe limit.



Insulation specification adopted

THICKNESS

50 mm

DENSITY

100 kg/m³

CLADDING

24g Al

Lux level measurement

The lighting assessment uses a calibrated digital lux meter, following the methodology recommended by the Bureau of Energy Efficiency.

Area dimensions give the floor area and Room Index. Illuminance is measured at the centre of each grid on the working plane. Installed Load Efficacy (ILE) is measured lux per watt per square metre; the Installed Load Efficacy Ratio (ILER) is measured ILE against the target for that space type.

ILER	ASSESSMENT
≥ 0.75	Satisfactory to good
0.51 - 0.74	Review suggested
0.375 - 0.50	Urgent action required
< 0.375	Immediate action required

Area-wise lux measurement summary

S.N.	LOCATION	AREA M ²	AVG LUX	MEASURED LUX/W/M ²	TARGET	ILER	ASSESSMENT
1	Dyeing hall	420.0	186.0	32.5	25.0	1.30	Satisfactory
2	Stenter line	310.0	142.0	21.0	25.0	0.84	Satisfactory
3	Quality lab	48.0	410.0	37.8	40.0	0.95	Satisfactory
4	Boiler house	96.0	78.0	8.3	20.0	0.42	Urgent
5	Store passage	64.0	52.0	4.0	15.0	0.26	Immediate

ASSESSMENT	NO. OF LOCATIONS
Satisfactory	3
Review	0
Urgent	1
Immediate	1

INSTRUMENTS

The following instruments were used to carry out this assessment.



S.N.	INSTRUMENT	MAKE	MODEL	QTY
1	3 Phase Energy Analyzer CLASS S	Krykard	CA 8336	1
2	Power Quality Analyzer	Krykard	ALM 35	1
3	Clamp on Power Meter	Fluke	345	1
4	Thermal Imaging Camera	FLIR	E8-XT	1
5	Infrared Thermometer	Fluke	62 MAX+	1